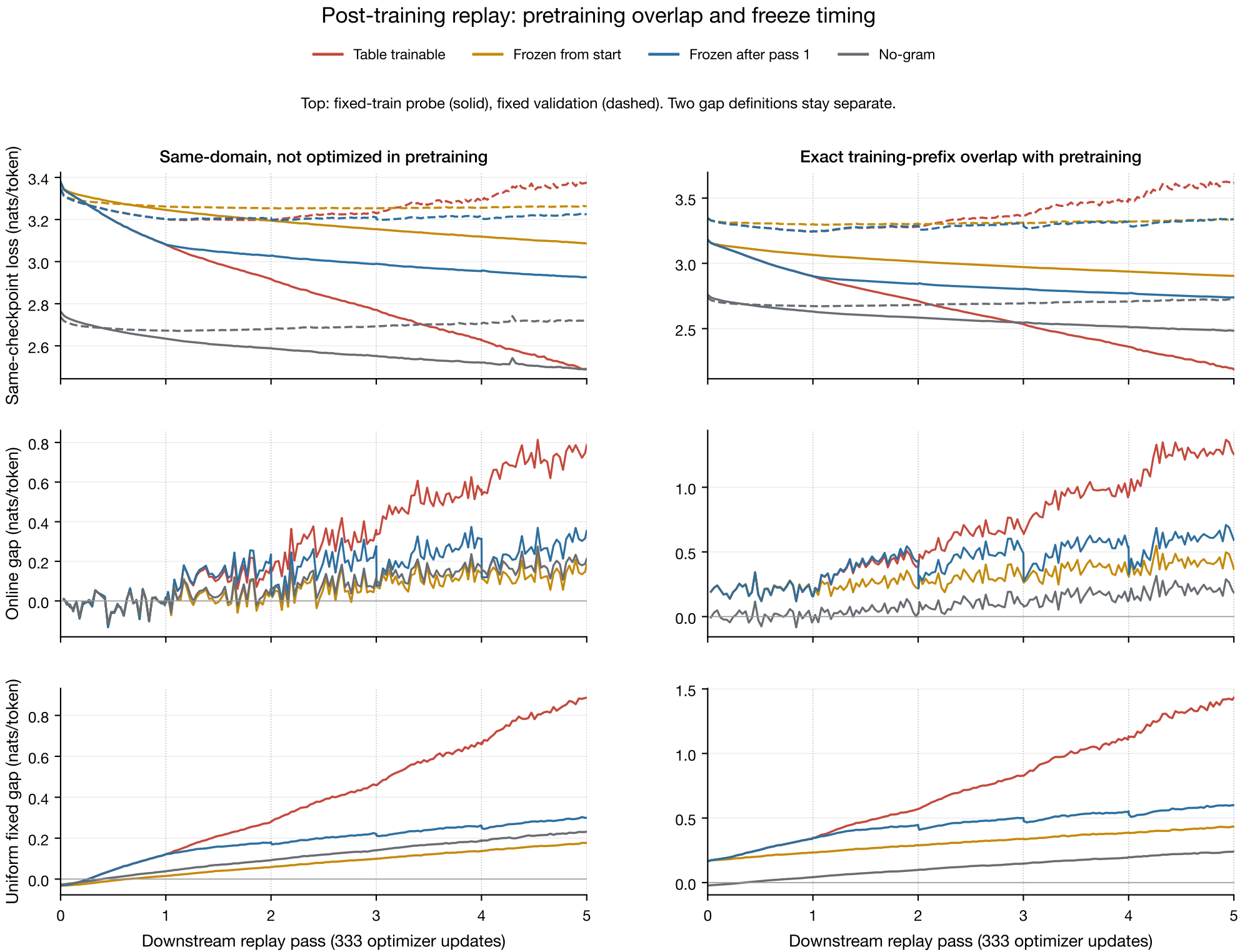

Post-training replay: pretraining overlap and freeze timing
Table trainable
Frozen from start
Frozen after pass 1
No-gram
Top: fixed-train probe (solid), fixed validation (dashed). Two gap definitions stay separate.
Same-domain, not optimized in pretraining
Exact training-prefix overlap with pretraining
Same-checkpoint loss (nats/token)
Online gap (nats/token)
Uniform fixed gap (nats/token)
Downstream replay pass (333 optimizer updates)
Downstream replay pass (333 optimizer updates)
Seed 42; input trigram R=2^20; LR=1.5e−4; table LR=128×; A100 microbatch 18 × accumulation 4.
Fixed probe: 288 rows across each exact pool. After-pass-1 freeze precedes update 334. No smoothing.